

Package ‘lwbgt’

October 1, 2026

Type Package

Title Reference-Compatible Outdoor Wet Bulb Globe Temperature

Version 1.0.1

Description Computes outdoor wet bulb globe temperature using the reference-compatible Liljegren numerical model described by Liljegren et al. (2008) <[doi:10.1080/15459620802310770](https://doi.org/10.1080/15459620802310770)>. Provides dependency-free, vectorized access to the native calculation with explicit input units, per-row validation, and deterministic failure reporting.

License Apache License (== 2.0)

Copyright See file inst/COPYRIGHTS.

URL <https://github.com/zyf0717/lwbgt>,
<https://zyf0717.r-universe.dev/lwbgt>

BugReports <https://github.com/zyf0717/lwbgt/issues>

Depends R (>= 4.3.0)

Encoding UTF-8

NeedsCompilation yes

Author Yifei Zheng [aut, cre, cph] (ORCID:
<<https://orcid.org/0009-0001-4858-2025>>),
James C. Liljegren [ctb],
Nels Larson [ctb],
UChicago Argonne, LLC [cph]

Maintainer Yifei Zheng <zyf0717@gmail.com>

Repository CRAN

Date/Publication 2026-10-01 15:50:02 UTC

Contents

lwbgt-package	2
calculate	2
esat	3
lwbgt_input	4

Index	6
-------	---

lwbgt-package	Reference-Compatible Outdoor Wet Bulb Globe Temperature
---------------	---

Description

Dependency-free, vectorized access to the Liljegren outdoor wet bulb globe temperature numerical model.

Details

Inputs use the same explicit field names and units as the native and Python interfaces. The package performs row-level validation and reports failures without aborting calculations for other rows.

References

Liljegren, J. C., Carhart, R. A., Lawday, P., Tschopp, S., and Sharp, R. (2008). Modeling the Wet Bulb Globe Temperature Using Standard Meteorological Measurements. *Journal of Occupational and Environmental Hygiene*, 5(10), 645–655. doi:[10.1080/15459620802310770](https://doi.org/10.1080/15459620802310770).

calculate	Calculate Outdoor Wet Bulb Globe Temperature
-----------	--

Description

Calculates one result per input row while preserving row order. Invalid rows do not prevent valid rows from being calculated.

Usage

```
calculate(input)
```

Arguments

input	A data frame containing the columns produced by lwbgt_input(). Additional columns are ignored.
-------	--

Value

A plain data frame containing integer status, character status_message, and five numeric result columns. Status zero indicates success; -1 indicates solver non-convergence; 1 indicates missing input; 2 indicates invalid input; and 3 indicates an invalid native result. Numeric outputs are NA for nonzero statuses.

Examples

```
input <- lwbgt_input(
  year = 2024, month = 4, day = 15, hour = 14, minute = 30,
  gmt_offset_hours = 8, averaging_minutes = 60, urban = 1,
  latitude_deg_north = 1.3521, longitude_deg_east = 103.8198,
  solar_w_m2 = 742, pressure_hpa = 1008.4,
  air_temperature_c = 32.1, relative_humidity_percent = 68,
  wind_speed_m_s = 2.8, wind_height_m = 10,
  vertical_temperature_difference_c = 1
)
calculate(input)
```

 esat

Calculate Saturation Vapour Pressure

Description

Calls the native Liljegren saturation-pressure calculation. Missing inputs propagate to NA; invalid non-missing inputs produce NA and one warning per call.

Usage

```
esat(temperature_k, phase = 0L)
```

Arguments

temperature_k Numeric temperatures in kelvin.

phase Zero for liquid water or one for ice.

Value

A numeric vector of saturation vapour pressures in hectopascals.

Examples

```
esat(c(263.15, 273.15, 303.15))
esat(263.15, phase = 1L)
```

lwbgt_input

Construct WBGT Input Records

Description

Constructs one or more input rows. All arguments are required; scalar values are recycled to the common non-scalar length.

Usage

```
lwbgt_input(
  year, month, day, hour, minute, gmt_offset_hours, averaging_minutes,
  urban, latitude_deg_north, longitude_deg_east, solar_w_m2,
  pressure_hpa, air_temperature_c, relative_humidity_percent,
  wind_speed_m_s, wind_height_m, vertical_temperature_difference_c
)
```

Arguments

year	Integer-valued Gregorian year from 1950 through 2049.
month, day, hour, minute	Integer-valued local date and time fields. A month of zero makes day a day-of-year value.
gmt_offset_hours	Local standard time minus GMT, in whole hours.
averaging_minutes	Input averaging interval, in minutes.
urban	Zero for rural wind scaling or one for urban wind scaling.
latitude_deg_north, longitude_deg_east	Coordinates in degrees north and degrees east.
solar_w_m2	Solar irradiance in watts per square metre.
pressure_hpa	Barometric pressure in hectopascals.
air_temperature_c	Dry-bulb air temperature in degrees Celsius.
relative_humidity_percent	Relative humidity in percent.
wind_speed_m_s	Wind speed in metres per second.
wind_height_m	Wind measurement height in metres.
vertical_temperature_difference_c	Upper-minus-lower temperature difference in degrees Celsius. Only whether it is < 0 or ≥ 0 is used. It can affect results only for rural nighttime records when <code>wind_height_m</code> differs from 2 m and <code>wind_speed_m_s</code> is below 2.5.

Value

A plain data frame with the 17 native input columns.

Examples

```
input <- lwbgt_input(  
  year = 2024, month = 4, day = 15, hour = 14, minute = 30,  
  gmt_offset_hours = 8, averaging_minutes = 60, urban = 1,  
  latitude_deg_north = 1.3521, longitude_deg_east = 103.8198,  
  solar_w_m2 = 742, pressure_hpa = 1008.4,  
  air_temperature_c = 32.1, relative_humidity_percent = 68,  
  wind_speed_m_s = 2.8, wind_height_m = 10,  
  vertical_temperature_difference_c = 1  
)
```

Index

calculate, [2](#)

esat, [3](#)

lwbgt (lwbgt-package), [2](#)

lwbgt-package, [2](#)

lwbgt_input, [4](#)